

Vincent Tong

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Perth, Australia | Open to relocate to Singapore

PROFESSIONAL SUMMARY

Backend-focused Software Engineer with industry and project experience building Go services, APIs, and data pipelines. Master of IT graduate with hands-on experience in PostgreSQL, AWS, and containerised deployment.

• Rate Pulse

December 2025 - Present

Software Engineer

Australia

- Designed and self-hosted a not-for-profit FX platform helping international users compare exchange rates, fees, taxes, historical trends, and market news.
- Developed backend **microservices** in **Go**, using **REST** and **gRPC** for client-server communication. Contributed to building and maintaining **PostgreSQL** databases. Used **Redis** to cache **frequently accessed, rarely changing data**, reducing **P95 latency** at **20 QPS** from **7–10 seconds** to approximately **250 ms**.
- Developed a multi-source scraping engine to collect and normalise structured market data using **Python**, **Selenium**, and **GitHub Actions**.
- Built **CI/CD** pipelines with **GitHub Actions**. Migrated services from **AWS** to a lower-cost cloud deployment on **OVHcloud**, using **Docker** and **Kubernetes**, reducing infrastructure costs.
- Built and deployed a responsive frontend using **Next.js**, and **TypeScript** on **Cloudflare**, with server-side rendering, parallel data fetching across backend services, and typed API responses to reduce integration errors.

• Lions Eye Institute

July 2025 – November 2025

Software Engineer Intern

Australia

- Developed an internal web application using **Python**, and **FastAPI**. Leveraged **embedded SQLite database**, to manage data efficiently, **minimise memory usage**, and **database network I/O overhead**, on a **low-resource server**. Reduced the review process from around **1 hour** to **approximately 10 minutes**.
- Developed an internal web using **Next.js**, **ASP.NET Core**, and **PostgreSQL** for managing laboratory workflows.

• Samsung Display

August 2023 – October 2023

Software Engineer Intern

Vietnam

- Developed an end-to-end desktop application using **.NET** and a **SQL Server** database to manage projects.
- Deployed an AI model for display-panel defect detection, reducing manual inspection time by **40%**.

• VNPT Group

October 2022 – June 2023

Data Engineer

Vietnam

- Developed and migrated large-scale data pipelines, processing approximately **100 million** data records daily, using **Pandas**, **Spark**, and **Hadoop**. Reduced the data pipeline runtime by **3 times**.
- Developed backend services for wrapping core **AI models**, using **Python**, **FastAPI**, and **PyTorch**.

CERTIFICATIONS & AWARDS

• AWS Certified Developer – Associate – Amazon Web Services

February 2026 – February 2029

• Out of the Box Idea Award – VAIT Hackathon 2025

March 2025

• Best Trainee Award – Samsung Display Vietnam

October 2023

PROJECTS

• GameOps Lab

July 2026 – Present

Go, AWS SAM, Lambda, API Gateway, EventBridge, SQS, DynamoDB, Docker

 

- Built an event-driven platform in Go that captured gaming sessions and generated weekly analytics through an AWS-compatible serverless architecture using API Gateway, Lambda, EventBridge, SQS, and DynamoDB.
- Implemented resilient asynchronous processing with retry policies, dead-letter queues, idempotency keys, and conditional writes to prevent message loss and duplicate analytics updates.
- Defined the infrastructure through AWS SAM and documented architectural trade-offs involving Lambda versus containers, DynamoDB versus PostgreSQL, least-privilege IAM, observability, cost, and migration from local emulation to production AWS.

SKILLS

- **Programming Languages:** Go, Python, C#, TypeScript/JavaScript, SQL
- **Backend & Web Technologies:** Gin, FastAPI, ASP.NET Core, REST APIs, gRPC, Next.js, React
- **Databases & Data Processing:** PostgreSQL, Redis, MongoDB, SQLite, PySpark, Hadoop
- **Cloud & DevOps:** AWS, Docker, Kubernetes, GitHub Actions, Git, Linux

EDUCATION

• The University of Western Australia

February 2024 – December 2025

Master of Information Technology, Software Systems Specialisation

Perth, Australia

• Hanoi University of Science and Technology

August 2019 – September 2023

Bachelor of Information Technology

Hanoi, Vietnam